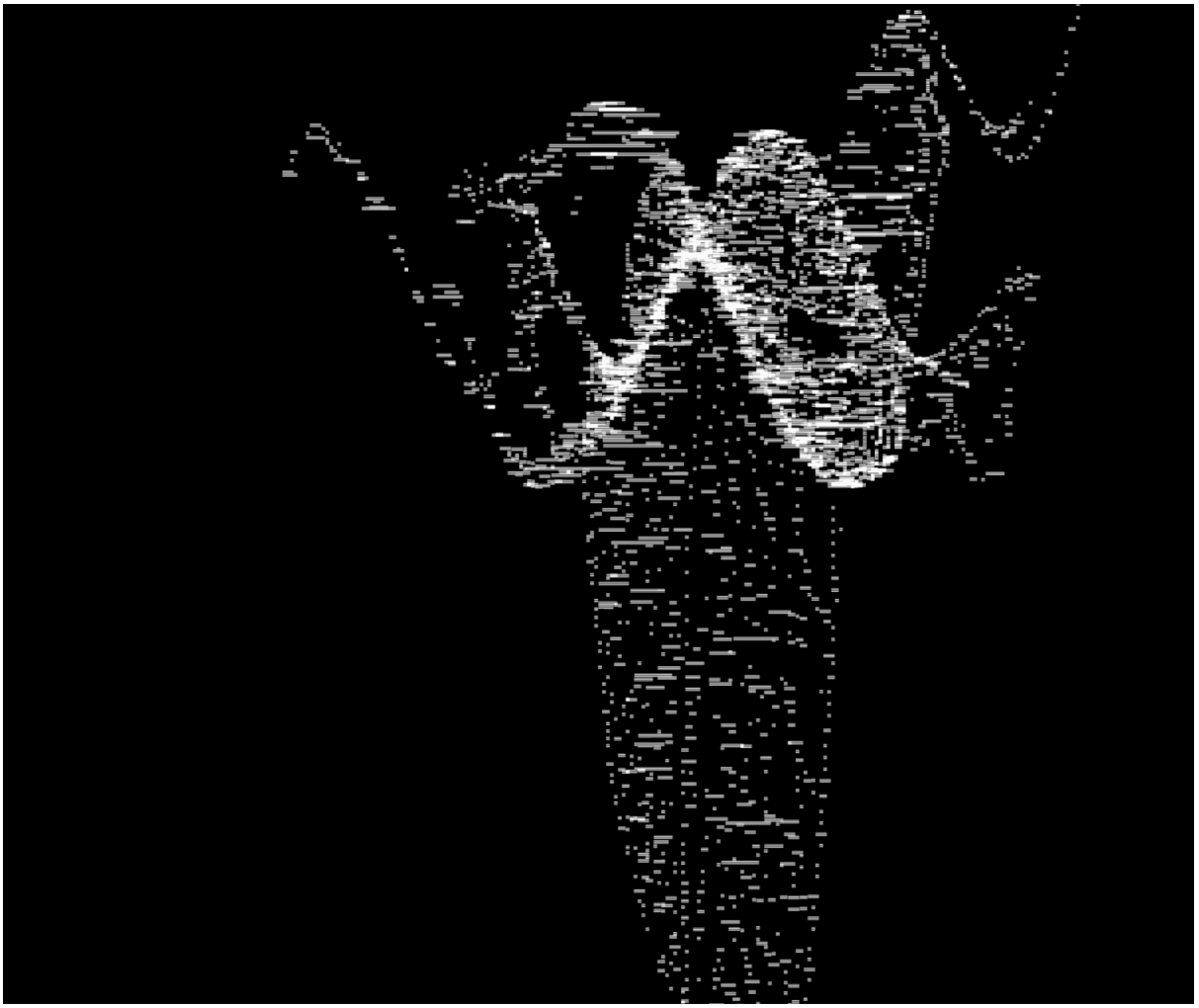
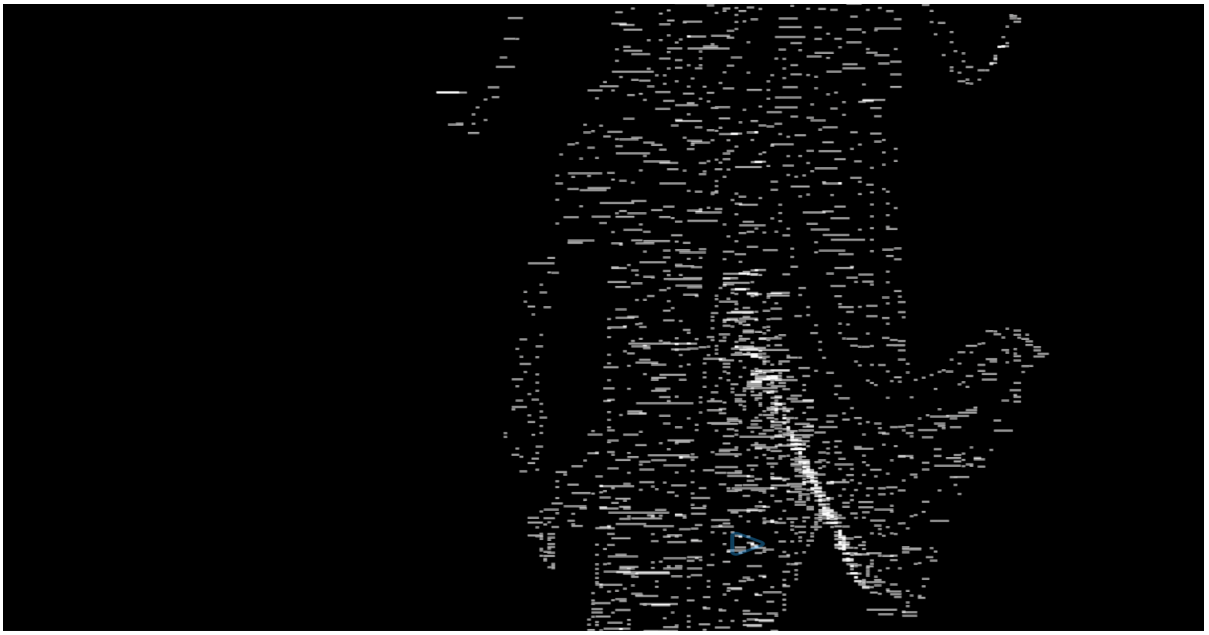
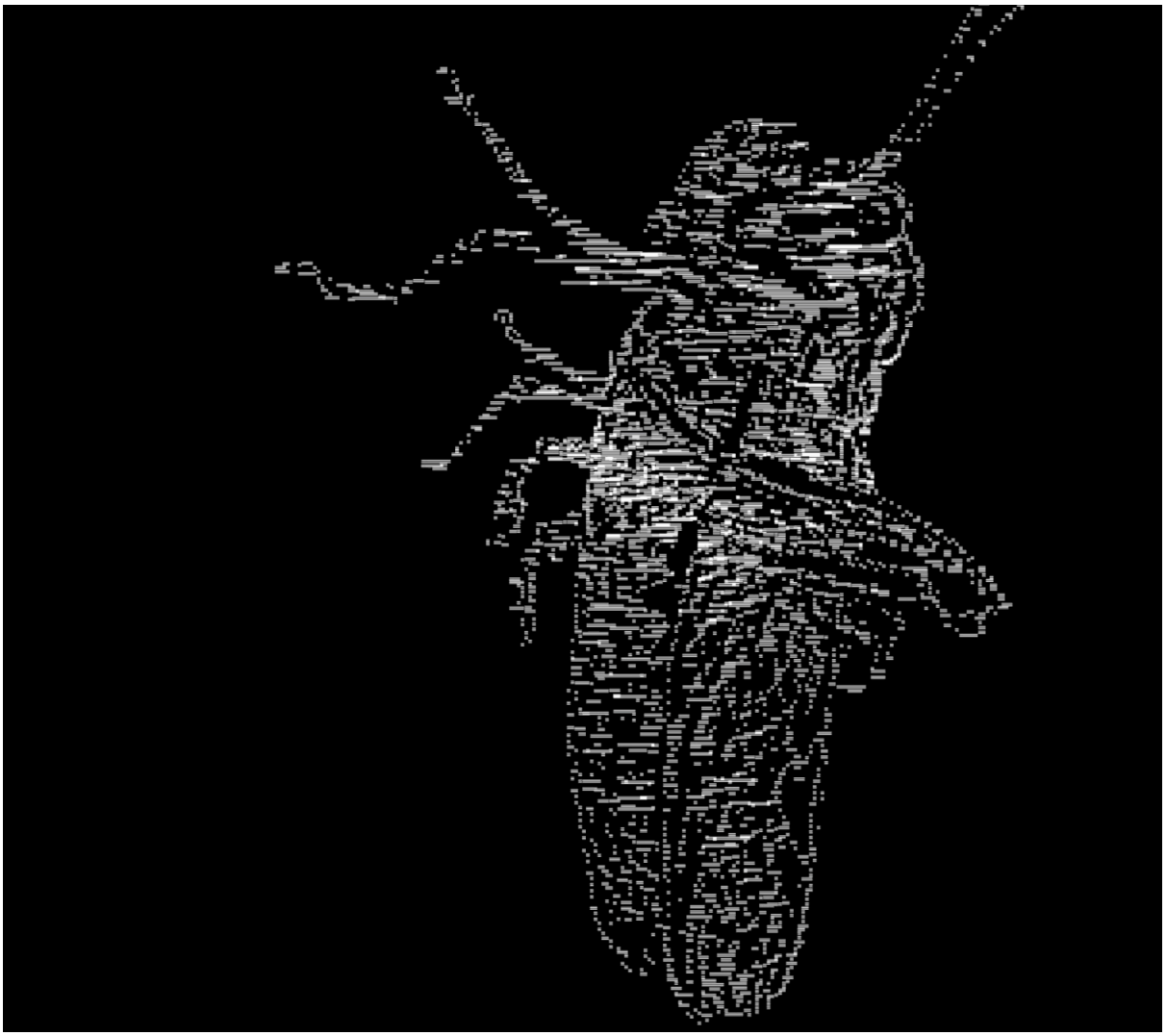
Project 2: locust.most (music.host)

Locust.most is a keyboard-playable synthesiser, the output simultaneously deforms the topology map of an ASCII-esc image of a locust. Tone.js provides the audio synthesis layer, built on top of the browser's Web Audio API. Audio is handled by Tone.js through a polyphonic sawtooth synthesiser, shaped by a lowpass filter, a reverb with a 3.5 second decay, and a hard limiter. A fast Fourier transform analyses the live audio energy every frame and feeds back into the visual motion for the physical volume and texture of the sound also modulates intensity of the (locust's) lines' movements. Three.js handles all WebGL rendering, the scene uses an OrthographicCamera to keep the "drawing" flat and print-like. The per-frame animation is done by directly mutating the underlying Float32Array rather than rebuilding geometry. Since WebGL does not reliably support line widths beyond one pixel on most (mine) GPU drivers, each segment is rendered as three stacked copies with a small vertical offset.

The audio reactivity models on vertexshaderart.com, with vertex positions calculated frame by frame, are the conceptual predecessor to the FFT-driven displacement system used in my project. Produced imagery through mathematical functions mapping an index to a position where no meshes, no scene graphs, no intermediate data structures could. The entire visual realm is implicit in complex formula. A piece of inspiration being *Antagonism* by kolargon is written within the constraints of vertexshaderart.com, a platform that provides only a GLSL vertex shader. Max Mitchell's *Unknown Lines* is the most direct technical precedent for this project. Built in Three.js and explicitly described as being inspired by the *Unknown Pleasures* album cover is another music visualiser in which only horizontal lines react to an existing audio track, deforming similarly to represent frequency and amplitude data.

The locust was chosen deliberately for in entomology and in cultural mythology, it is the insect of transformation under duress (depending on the interpertation). When music is played through the keyboard, the linear rows of the image bulge and lift as if the topology map has come under geological pressure. Of course, a great inspiration for this project being the same contour-map grammar as the *Unknown Pleasures* cover. At rest, these lines faithfully reconstruct the insect's form and the locust exists as implication: the gaps between lines, the shape made by density and absence. As life progresses in ways beyond comprehension, or at least cannot verbally align directly with certain untold struggles has furthered my reasons as to why I chose an almost homage to their (Joy Division's) masterpiece.

https://github.com/consumedlocusts/CART263AA/tree/main/locust.most

https://consumedlocusts.github.io/CART263AA/locust.most/